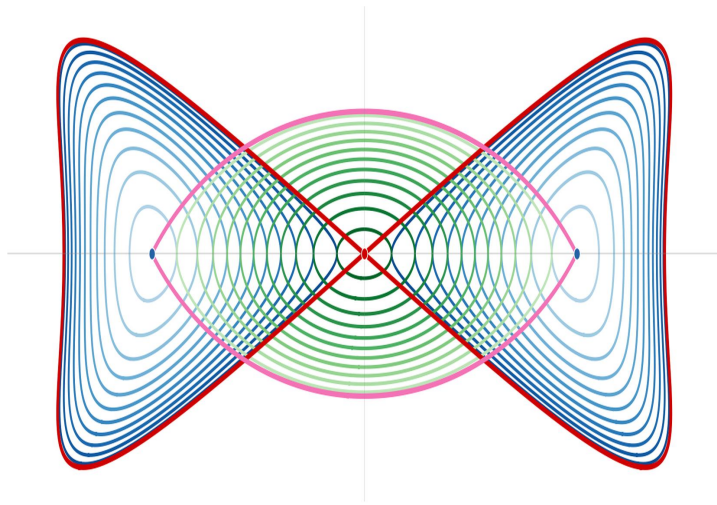
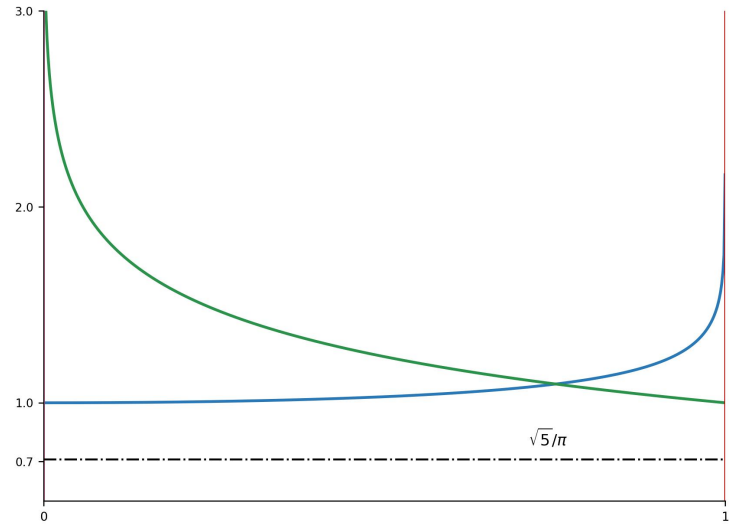




(a) Real and Complex cycles, by Abel–Wick transform.



(b) Periods and Abel's identity.

The μ-stratum	$2H_\mu = p^2 + q^2 + \frac{1}{2}(q^3 - 3p^2q) + \mu(q^2 - 3p^2)^2, \quad \alpha = 2H_\mu.$
Transform	$2H_\mu(p, -4 - q) - 2H_\mu(p, q) = -(16\mu - 1)(q + 2)(3p^2 - q^2 - 4q - 8),$ $q \mapsto -4 - q \implies \mu = \frac{1}{16}, \quad (p, q) = (0, -2), \quad \alpha = 1.$
Curve	$\alpha = 2H = p^2 + q^2 + \frac{1}{2}(q^3 - 3p^2q) + \frac{(q^2 - 3p^2)^2}{16}, \quad \rho = (2H)_p = \frac{p}{4}(9p^2 - 3(q + 2)^2 + 20).$
Period form	$\omega = \frac{2 \, dq}{\rho} = \frac{dq}{H_p}.$
Real period	$T_1(\alpha) = \frac{1}{2\pi} \oint_{\gamma_\alpha^{(1)}} \omega, \quad T_1(0) = 1, \quad \gamma_\alpha^{(1)} \subset (p, q) \text{ around } (0, 0).$
Complex period	$T_2(\alpha) = \frac{\sqrt{5}}{4\pi} \oint_{\gamma_\alpha^{(2)}} i\omega, \quad T_2(1) = 1, \quad \gamma_\alpha^{(2)} \subset (ip, q) \text{ around } (0, -2).$

Annihilator	$A_\alpha = P_0 + P_1 \partial_\alpha + P_2 \partial_\alpha^2, \quad P_0 = \frac{3}{4}\alpha(9\alpha - 64),$ $P_1 = 54\alpha^3 - 195\alpha^2 - 112\alpha + 128, \quad P_2 = \alpha(\alpha - 1)(3\alpha - 8)(9\alpha + 16).$
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Factorization	$P_1 = P'_2 - \frac{6}{3\alpha - 8}P_2, \quad A_\alpha = (3\alpha - 8)^2 \partial_\alpha \left(\frac{P_2}{(3\alpha - 8)^2} \partial_\alpha \right) + P_0.$
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Abel's identity	$\frac{\alpha(\alpha - 1)(9\alpha + 16)}{3\alpha - 8} (T_2 T'_1 - T_1 T'_2) = \frac{\sqrt{5}}{\pi}.$
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Certificate	Set $x = q + 2$. Then $\Xi = \frac{V(\alpha, p, x)}{8\rho^3}, \quad A_\alpha \circ \omega = d\Xi$, where $V = x \left[27\alpha^3(9p^2 - 3x^2 + 80) + 3\alpha^2(189p^2x^2 - 270p^2 - 27x^4 + 126x^2 + 280) \right.$ $\left. + 2\alpha(-27p^2x^4 - 648p^2x^2 + 4008p^2 + 9x^6 - 156x^4 + 456x^2 + 1120) \right.$ $\left. + 8(3x^2 - 16)(-9p^2x^2 + 108p^2 + 3x^4 - 32x^2 + 80) \right].$
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Pullback data	$D = (16 + 9\alpha^2)^{1/4}, \quad X = -\frac{108\alpha^2(\alpha - 1)(9\alpha + 16)^2}{(16 + 9\alpha^2)^3}, \quad \sigma = \frac{(3\alpha + 2)(32 - 48\alpha - 9\alpha^2)}{(16 + 9\alpha^2)^{3/2}}, \quad \sigma^2 = 1 - X.$
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$$D_2 = (25 - 18\beta + 9\beta^2)^{1/4}, \quad X_2 = \frac{108\beta(1 - \beta)^2(25 - 9\beta^2)}{(25 - 18\beta + 9\beta^2)^3}, \quad \beta = 1 - \alpha.$$

Local periods	$T_1(\alpha) = \frac{2}{D} {}_2F_1\left(\frac{1}{12}, \frac{5}{12}; 1; X\right), \quad T_2(1 - \beta) = \frac{\sqrt{5}}{D_2} {}_2F_1\left(\frac{1}{12}, \frac{5}{12}; 1; X_2\right).$
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Series	$T_1(16x) = 1 + 24x^2 + 120x^3 + 2520x^4 + 25200x^5 + 397320x^6 + 5045040x^7 + \dots,$ $T_2(1 - 400x) = 1 + 132x + 29220x^2 + 7816080x^3 + 2324945700x^4 + 738712234512x^5 + \dots.$
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Global basis	$\psi_0(\alpha) = \frac{2\sqrt{\pi}}{D \Gamma(11/12) \Gamma(7/12)} {}_2F_1\left(\frac{1}{12}, \frac{5}{12}; \frac{1}{2}; 1 - X\right),$ $\psi_1(\alpha) = -\frac{4\sqrt{\pi} \sigma}{D \Gamma(1/12) \Gamma(5/12)} {}_2F_1\left(\frac{11}{12}, \frac{7}{12}; \frac{3}{2}; 1 - X\right).$
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Global periods	$T_1(\alpha) = \psi_0(\alpha) + \psi_1(\alpha), \quad T_2(\alpha) = \frac{\sqrt{5}}{2}(\psi_0(\alpha) - \psi_1(\alpha)).$
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Birational maps $u = \frac{q - \sqrt{3}p}{2}, \quad v = \frac{q + \sqrt{3}p}{2}, \quad A(u) = \frac{3u^2 + 6u + 4}{12}, \quad B(u) = \frac{u(3u + 2)}{6},$
 $x = u + 1, \quad y = 2A(u)v + B(u),$
 $q = x - 1 + \frac{y - B(x - 1)}{2A(x - 1)}, \quad p = \frac{1}{\sqrt{3}} \left[\frac{y - B(x - 1)}{2A(x - 1)} - (x - 1) \right].$

Jacobi quartic $12y^2 = -x^4 + (2 + 3\alpha)x^2 + \alpha - 1, \quad \omega = \frac{dx}{\sqrt{3}y}.$

Quartic period $T_1(\alpha) = \sqrt{\frac{2}{2 + 3\alpha}} {}_2F_1\left(\frac{1}{4}, \frac{3}{4}; 1; z_4\right), \quad z_4 = \frac{\alpha(9\alpha + 16)}{(3\alpha + 2)^2}.$

Birational again $z_4 = \frac{4m}{(1 + m)^2}, \quad a^2, b^2 = \frac{2 + 3\alpha \pm \sqrt{\alpha(16 + 9\alpha)}}{2}, \quad k = \frac{a - b}{a + b}, \quad m = k^2.$
 $u = k \frac{x + b}{x - b}, \quad v = \frac{\sqrt{3}(u - k)^2 y}{b(a - b)},$
 $x = b \frac{u + k}{u - k}, \quad y = \frac{b(a - b)}{\sqrt{3}(u - k)^2} v.$

Legendre cubic $v^2 = u(u - 1)(u - m), \quad \omega = -\frac{2}{a + b} \frac{du}{v}.$

Cubic period $T_1(\alpha) = \frac{2}{a + b} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; m\right) = \frac{4K(k)}{\pi(a + b)}.$

Three periods $T_1(\alpha) = \sqrt{\frac{2}{2 + 3\alpha}} {}_2F_1\left(\frac{1}{4}, \frac{3}{4}; 1; z_4\right) = \frac{2}{a + b} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; m\right) = \frac{4K(k)}{\pi(a + b)}.$

Modular data $\tau(\alpha) = i \frac{K(k')}{K(k)}, \quad X = \frac{1728}{j(\tau)}.$

Parameter chain $m = k^2 = \frac{3\alpha + 2 - 2\sqrt{1 - \alpha}}{3\alpha + 2 + 2\sqrt{1 - \alpha}}, \quad X = \frac{27m^2(1 - m)^2}{4(1 - m + m^2)^3} = X(\alpha).$

j-line solution $\mathcal{E}_4(\alpha) = \frac{D}{2} T_1(\alpha) = {}_2F_1\left(\frac{1}{12}, \frac{5}{12}; 1; X\right),$
 $\mathcal{L}_X = X(1 - X)\partial_X^2 + \left(1 - \frac{3}{2}X\right)\partial_X - \frac{5}{144},$
 $\mathcal{L}_X \mathcal{E}_4 = \mathcal{L}_X E_4(\tau)^{1/4} = 0, \quad \mathcal{E}_4(0) = E_4(i\infty)^{1/4} = 1.$

Fourth power $E_4(\tau(\alpha)) = \mathcal{E}_4(\alpha)^4 = \left(1 + \frac{9\alpha^2}{16}\right) T_1(\alpha)^4 = \left(1 + \frac{9\alpha^2}{16}\right) \left(\frac{4K(k)}{\pi(a + b)}\right)^4.$

Algebraic factor $(a + b)^4(1 - m + m^2) = 16 + 9\alpha^2, \quad 1 - m + m^2 = 1 - k^2 k'^2.$

Ramanujan $E_4(\tau) = \left(\frac{2K(k)}{\pi}\right)^4 (1 - k^2 k'^2).$

“9. We may get a still closer approximation from the following results. It is known that

$$1 + 240 \sum_{r=1}^{\infty} \frac{r^3 q^{2r}}{1 - q^{2r}} = \left(\frac{2K}{\pi}\right)^4 (1 - k^2 k'^2), \quad \dots”$$

— S. Ramanujan, *Modular equations and approximations to π* , §9 (1914).